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GitHub Repository: https://github.com/chibunom-netizen/Investigating-the-impact-of-hidden-layer-MLP/tree/main

Investigating the impact of hidden layer depth on the performance and overfitting behaviour of Multilayer Perceptron's

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1. Introduction

Neural networks are a powerful way for computers to learn patterns in data, especially when those patterns are complex or non-linear. Among the most popular neural networks are the Multilayer Perceptron's (MLPs). In fact, the MLPs are the building blocks for many other popular deep learning architectures. Despite the relative simplicity of the MLP, this model is widely used for classification and regression tasks, making it a must-know for anyone interested in deep learning architectures.

Neural networks are used in a variety of real-world applications, from image and speech recognition, fraud detection, and even recommendation systems. Their ability to learn from large amounts of data and reveal meaningful patterns in the process makes them a core component of contemporary artificial intelligence systems. As training datasets grow larger and more complex, it becomes increasingly important for neural network designs to be effective. One key decision when building a Multilayer Perceptron (MLP) is choosing how many hidden layers to include. This choice matters because the number of hidden layers directly affects how well the network can learn and represent complex patterns in the data. Although a higher number of hidden layers makes the network capable of recognizing complex patterns, it also makes the network prone to overfitting, as well as complicates the training process.

This tutorial investigates the impact of the number of hidden layers in a Multilayer Perceptron (MLP) network on the performance of the network, as well as the training process. In addition, this tutorial seeks to understand how the network can be regularized using the early stopping technique to prevent overfitting in the network.

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2. Background

2.1 Multilayer Perceptron (MLP)

Multilayer perceptron is a feed-forward artificial neural network with three layers: the input layer, the hidden layer, and the output layer. In the MLP, there are several neurons in each layer, and the neurons in the current layer are connected to the next layer.

Each neuron in the MLP has the capacity to perform a simple operation, which is the weighted sum of the inputs and the bias. This output of the neuron is then passed through the activation function. One of the activation functions is the ReLU function, given by the equation:

- Output = max(0, input)

The activation function has the capacity to add non-linearity to the MLP, which is essential in the ability of the MLP to solve complex problems.

The MLP can be trained using backpropagation algorithm, combined with an optimization algorithm such as gradient descent or Adam. In the backpropagation algorithm, the MLP will make predictions on the data and then compare the output with the actual output using the loss function. This will give us the error, and then the error will be backpropagated through the MLP to give the minimum loss.

2.2 Model Depth and Learning Capacity

The depth of an MLP is measured by the number of hidden layers in the network. This directly affects its learning capacity. The capacity of a model is its ability to learn a wide range of functions.

- For simple functions, shallow networks (with one hidden layer) can approximate simple functions; they can capture simple relations in the data

- For complex functions, deeper networks (with more hidden layers) can approximate complex functions by learning complex relations in the data

For example, in image classification problems, simple relations in the data might include edges in an image, while complex relations might include shapes in an image.

However, there are some problems that arise as the depth of an MLP is increased:

- **Higher computational cost** due to increased parameters and operations

- **Longer training times**, as more layers require more update

- **Possibility of overfitting is increased,** as the model may learn noise rather than general patterns

Additionally, deeper networks can suffer from optimisation challenges such as vanishing gradients, where the gradient values become very small during backpropagation, making it difficult for earlier layers to learn effectively.

2.3 Overfitting and Early Stopping

Overfitting is when the model performs well on the training data but fails to generalise well on new data. This is due to the model learning the noise in the data instead of the underlying data structure.

Deep neural networks are prone to overfitting due to their capacity. They can learn complex data structures but are also able to memorise the data if not regularised.

Early stopping is a common technique used in the prevention of overfitting in models. A portion of the data is set aside during the training process, referred to as the validation set. The performance of the model on the validation set is then monitored. If the performance on the validation set has not improved in a certain number of iterations, the training process is stopped.

This helps in the prevention of the model learning the noise in the data and thus helps in the maintenance of the performance of the model on new data.

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3. Methodology

3.1 Objective

The objective of this experiment is to determine the effect of the number of hidden layers on the following factors:

- Accuracy
- Training time
- Learning behaviour
- Overfitting

Additionally, the experiment also uses early stopping to determine the effectiveness of this technique in improving the performance of the models and preventing overtraining.

3.2 Dataset

The experiment uses the sklearn digits dataset, which consists of.

- 8×8 Pixel images
- 10 classes – digit recognition (0–9)
- Each image is represented as a vector of numerical pixel values

This dataset is appropriate for this experiment because it is relatively simple and easy to process.

3.3 Model Configurations

Th experiment uses three different MLP models with different configurations:

Model	Hidden Layers
Model A	(100)
Model B	(100, 100, 100)
Model C	(100, 100, 100, 100, 100)

In all the models, the hidden layers have 100 units.

In addition, the experiment also uses a model with early stopping.

Model	Configuration
Model D	5 layers + early stopping

3.4 Evaluation Metrics

The experiment uses the following metrics to evaluate the performance of the models.

- **Test Accuracy**: This is the accuracy of the model on the test set.
- **Training Accuracy**: This is the accuracy of the model on the training set.
- **Training Time**: This is the time it takes to train the model.
- **Loss Curves**: This gives us the learning behaviour of the model.

The experiment uses all these metrics to give us a complete picture of the performance of the models.

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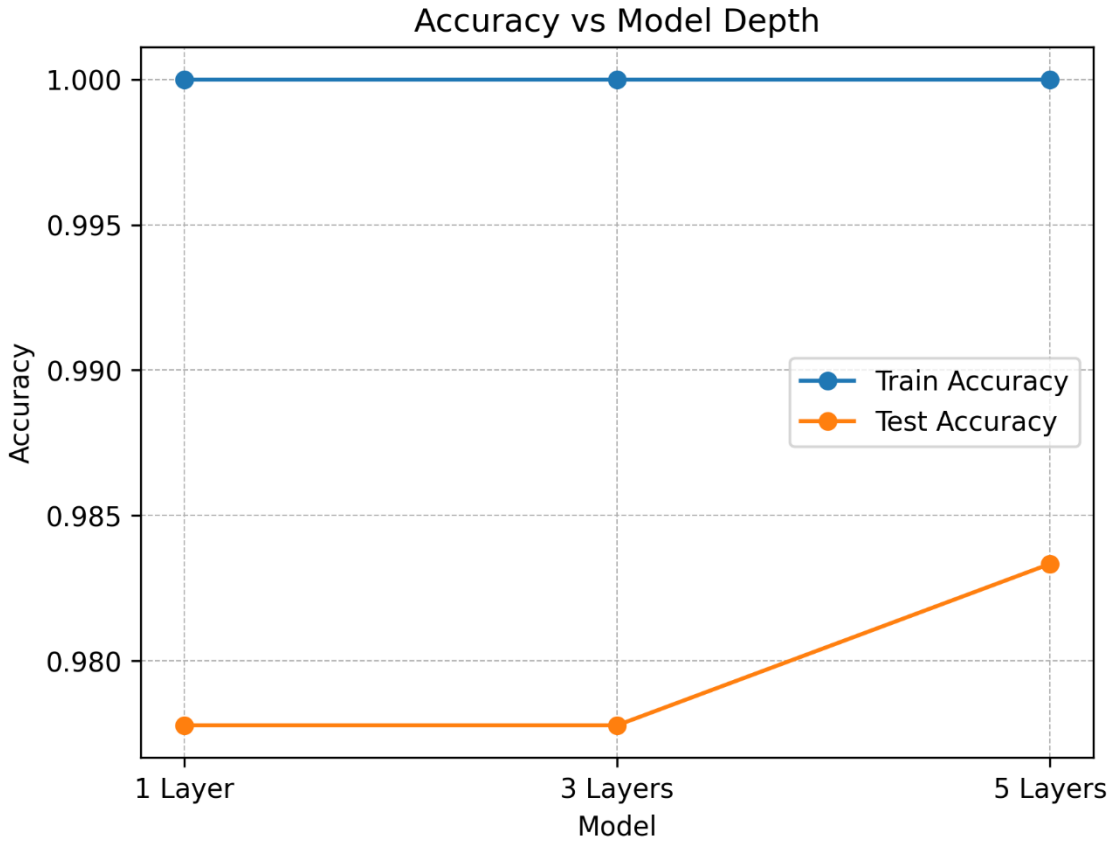
4. Results

4.1 Accuracy and Overfitting Analysis

The results indicate that the more the number of hidden layers in the model, the better the better the performance. This is true up to a point. It is evident that the 3-layer model has the best accuracy in terms of balancing training and test accuracy. This implies that the model can learn and generalise without overfitting.

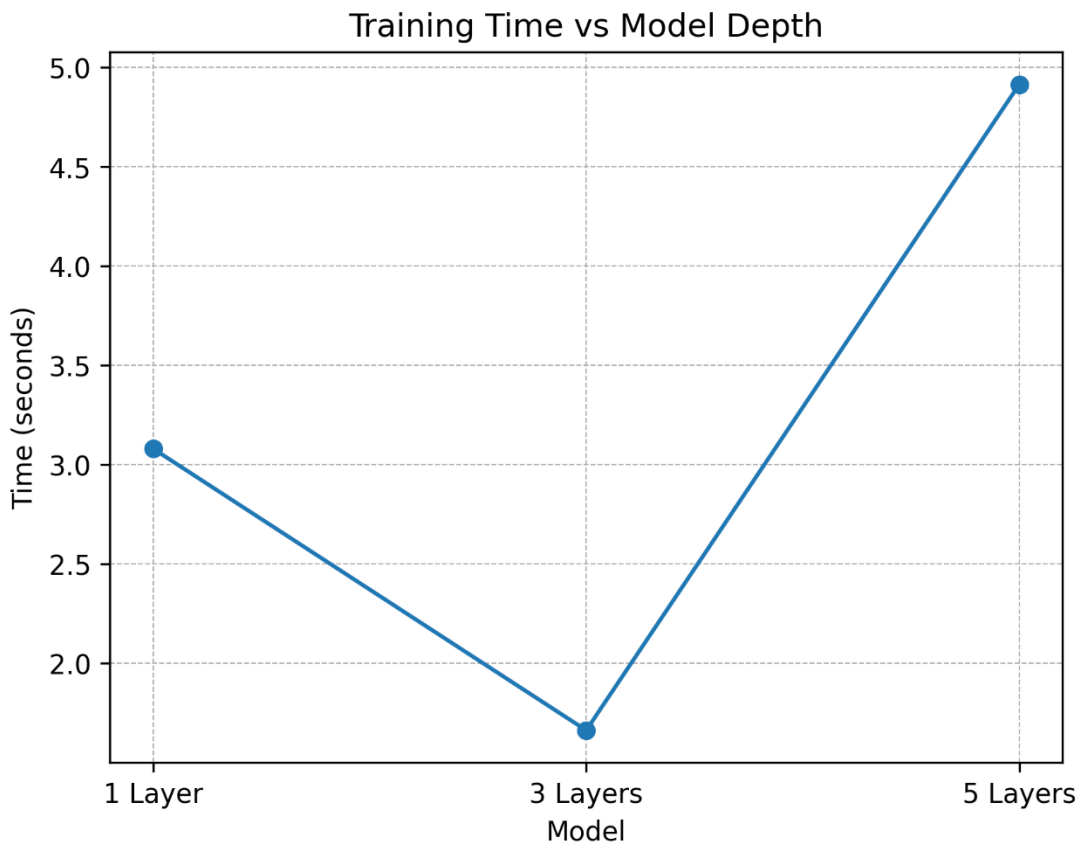
In contrast, the accuracy of the 5-layer model is high compared to the others. However, it does not show significant improvement in terms of accuracy. This implies that the model is already overfitting.



The increasing gap between training and test accuracy as depth increases provides clear evidence of overfitting in deeper models.

4.2 Training Time Comparison

Training time increases significantly as the number of hidden layers increases. This is expected, as deeper networks require more computations during both forward and backward propagation.

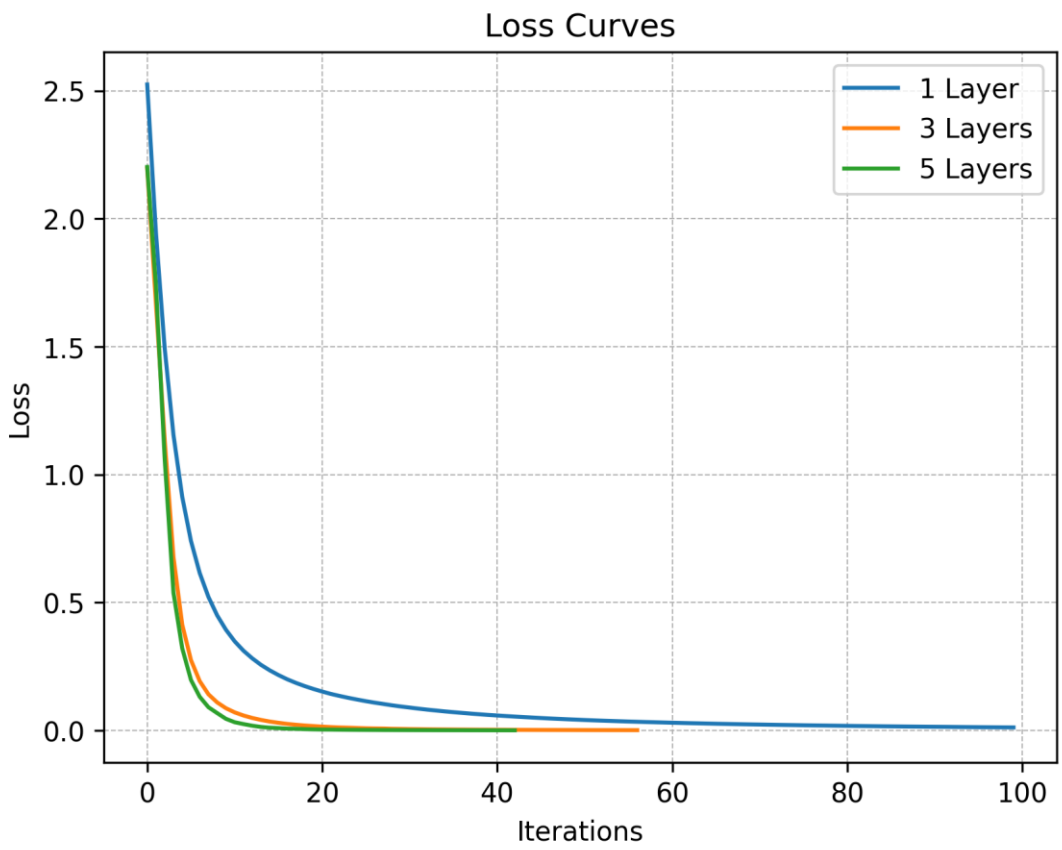


The results highlight an important trade-off: while deeper models have slightly higher accuracy than shallow models, they require more computational resources.

4.3 Learning Behaviour (Loss Curves)

Loss curves provide valuable insight into how the models learn over time.

- Shallow model – fast convergence to a minimum
- Deeper model – slower convergence
- Deepest model – fluctuations in loss curve

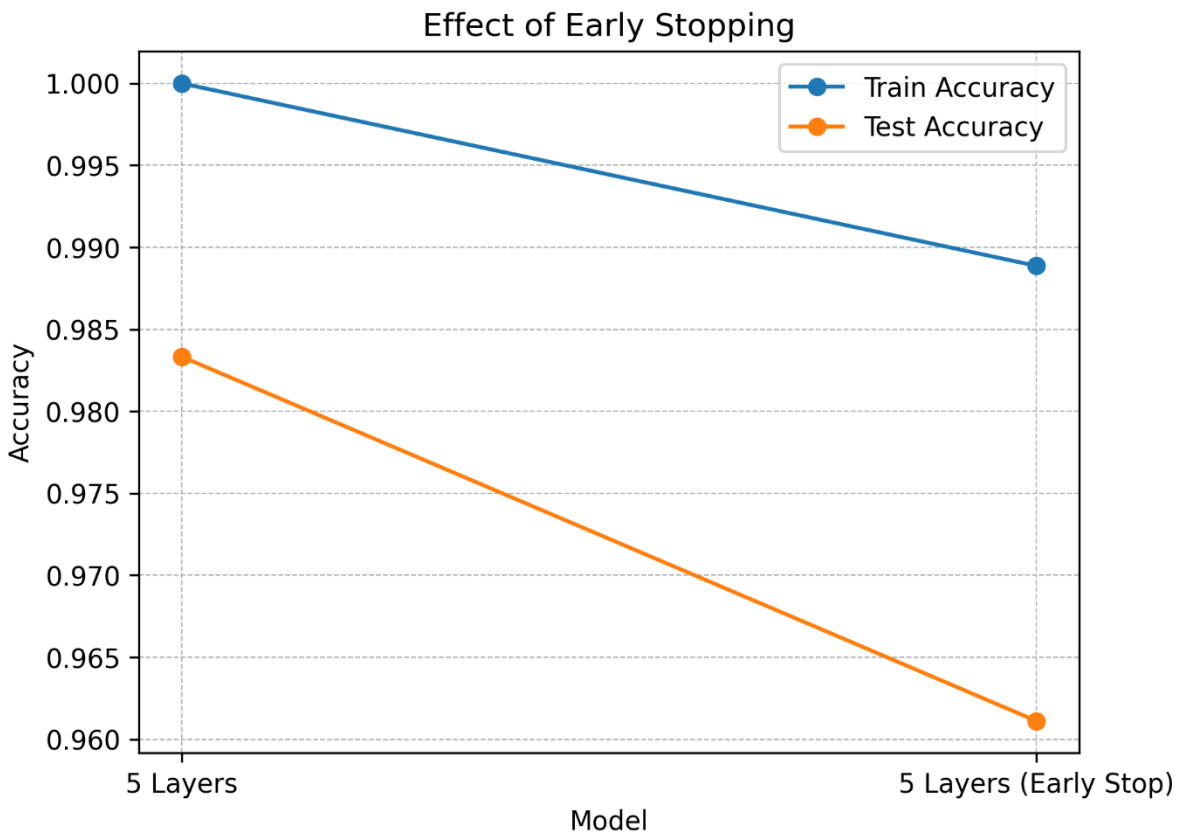


This shows that it is more difficult to train deeper models and that additional tuning is required.

4.4 Effect of Early Stopping

The early stopping model has better generalisation compared to the basic model that has 5 layers.

- Training is automatically stopped when the model fails to improve
- Overfitting is reduced
- Test accuracy is more stable
- Training time is reduced



This proves that early stopping is an efficient technique in the prevention of overfitting and the enhancement of efficiency in deep neural networks.

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5. Discussion

5.1 Key Findings

The experiment reveals several important insights:

- Adding hidden layers increases the capacity and performance of the model initially
- The performance increases are not significant beyond a certain level of depth
- Training time increases significantly with depth
- Deeper models are more prone to overfitting

5.2 Impact of Early Stopping

Early stopping is an important factor in the enhancement of the performance of the model. This technique helps in the prevention of overfitting of the model.

In the context of deep models, the risk of overfitting is higher due to the increased complexity of the model.

5.3 Theoretical Insights

The results correspond with theoretical expectations. Deeper networks provide increased representational power by applying multiple layers of non-linear transformations. However, this increased power also leads increases the variance of the network, which makes it more sensitive to noise.

Furthermore, optimisation challenges such as vanishing gradients and unstable convergence can make deeper networks harder to train effectively.

5.4 Practical Implications

From a practical standpoint:

- A moderate number of layers (2–3) is often sufficient for many tasks
- Increasing depth should be justified by the complexity of the problem
- Regularisation techniques such as early stopping are essential for deeper networks
- Efficient model design requires balancing performance, complexity, and computational cost

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6. Conclusion

The aim of this tutorial was to examine the effect of hidden layers on the performance of a Multilayer Perceptron.

The results have shown that increasing the depth of the network enhances its performance but only up to a certain level. After that, the performance improvement slows down. Deeper networks require more computational resources and are more prone to overfitting, making them less efficient in some cases.

The introduction of early stopping shows that regularisation techniques can significantly improve generalisation and reduce unnecessary training. This highlights the importance of not only selecting an appropriate model architecture but also to incorporate techniques to prevent overfitting.

Overall, the findings have shown that the appropriate design of a neural network is not necessarily based on making it more complex, but about finding the right balance between model capacity, generalisation, and efficiency.

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7. References

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- Scikit-learn Documentation
- LeCun, Y. et al. (1998). Gradient-based learning applied to document recognition